

PM1 Source Localization Using Lagrangian Stochastic Particle Dispersion Model

July 4, 2025 Fireworks — Lake Union, Seattle, WA

Estimated Source Location: 47.6389°N, 122.3352°W

All parameters derived from sensor observations and KSEA airport wind data.
No ground truth was used in parameter selection or source estimation.

1. Objective

This report describes the methodology used to estimate the location of a PM1 emission source (fireworks launch site) at Lake Union, Seattle, using concentration data from five low-cost PM1 sensors and meteorological wind observations from KSEA airport. The method uses a backward Lagrangian stochastic particle dispersion model to trace observed PM1 concentrations back to their origin. The source location is extracted as the concentration-weighted centroid of the top 2% of the resulting trajectory density field.

2. Input Data

2.1 PM1 Sensor Network

Six PM1 sensors were deployed around the south end of Lake Union. Each recorded PM1 concentration at approximately 1 Hz (~1.111 seconds per sample) for approximately 4 hours starting at 20:00 PDT on July 4, 2025. Sensor S4 was excluded from all analyses because it recorded anomalous concentrations (peak ~545 ug/m³ net, 50-100x higher than other sensors at similar distances), attributed to proximal contamination from a barbecue and consumer fireworks near the sensor site.

Sensor	Address Reference	Lat (N)	Lon (W)	Net PM1 (ug/m ³)
S1	2031 Fairview Ave E	47.6378	122.3295	4.3

S2	2838 Fairview Ave E	47.6469	122.3263	7.1
S3	2199 N Northlake Way	47.6493	122.3316	11.7
S5	1200 Westlake Ave N	47.6299	122.3394	6.6
S6	809 Fairview Pl N	47.6267	122.3353	4.7

Table 1. PM1 sensor locations and event-average net concentrations. Net PM1 is the mean over the analysis window (counter 7000-12000) minus background (counter 1000-5000).

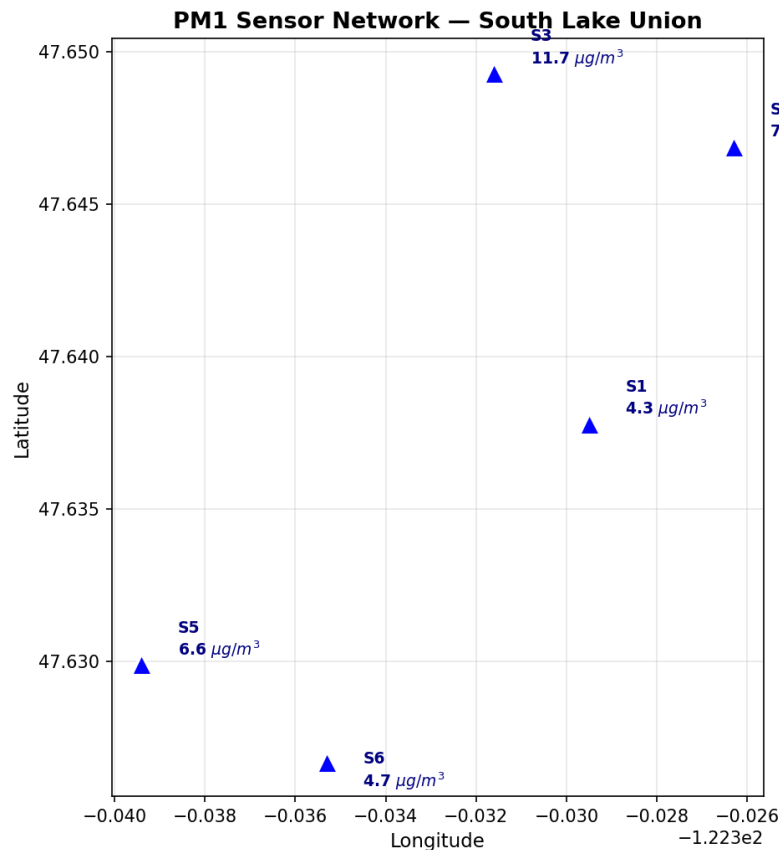


Figure 1. PM1 sensor network around south Lake Union, with event-average net PM1 concentrations. S3 (due north) recorded the highest concentration, indicating the plume traveled primarily northward.

2.2 PM1 Observations

Background PM1 was computed from each sensor's pre-fireworks period (counter 1000-5000, approximately 20:18-21:32 PDT). Net PM1 equals the observed concentration minus background, floored at $0.1 \mu\text{g}/\text{m}^3$. The fireworks produced a clear PM1 signal beginning at counter ~6500 (approximately 22:00 PDT) and persisting through counter ~12500 (approximately 23:30 PDT).

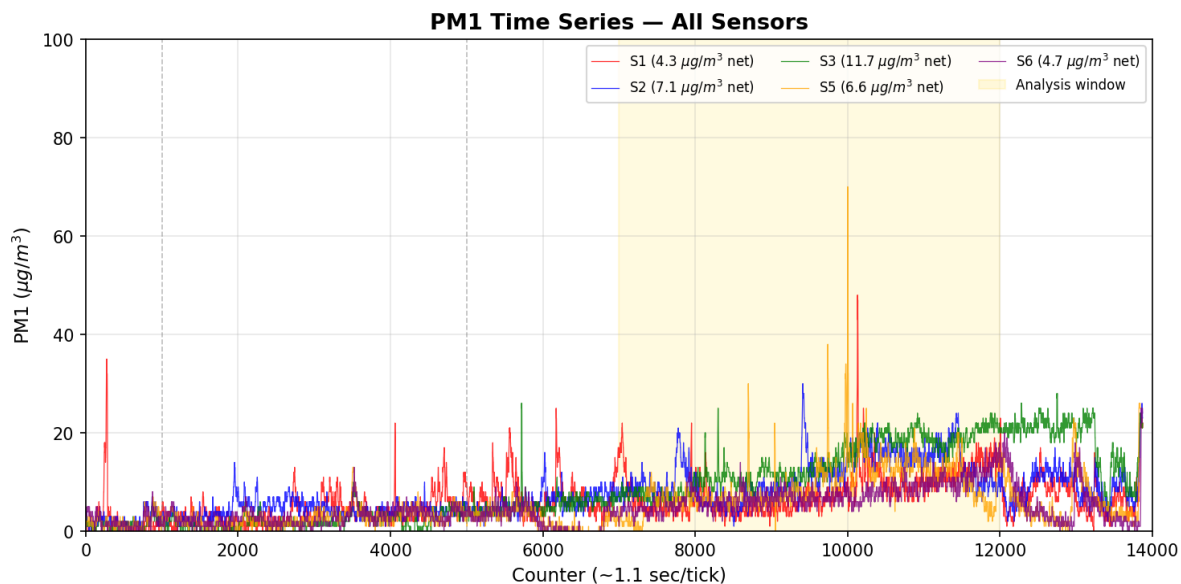


Figure 2. PM1 time series from all five sensors. Yellow shading marks the analysis window. S3 consistently records the highest concentrations, confirming northward plume transport.

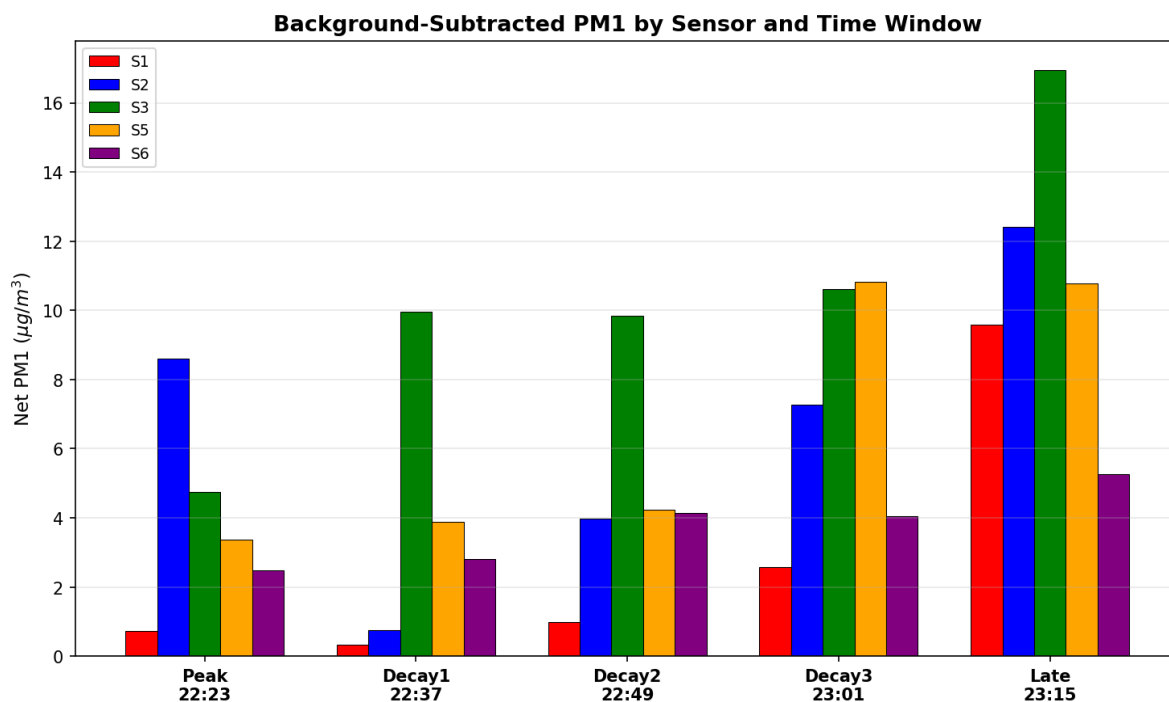


Figure 3. Background-subtracted PM1 by sensor and time window. The concentration pattern shifts progressively from S1 toward S3/S5 over time as the plume advects.

2.3 Wind Data

Wind observations were obtained from KSEA (Seattle-Tacoma International Airport) via the MesoWest METAR network. KSEA is located 21.5 km south of Lake Union at 47.4447 N, 122.3136 W, on flat open airport tarmac. The data has 5-minute temporal resolution, with wind speed in m/s and direction in degrees using the meteorological "FROM" convention. During the fireworks event

(21:00-23:30 PDT), KSEA reported a steady southwesterly wind: mean direction FROM 222 degrees, mean speed 3.4 m/s. No corrections were applied to the KSEA wind data; it was used as-is.

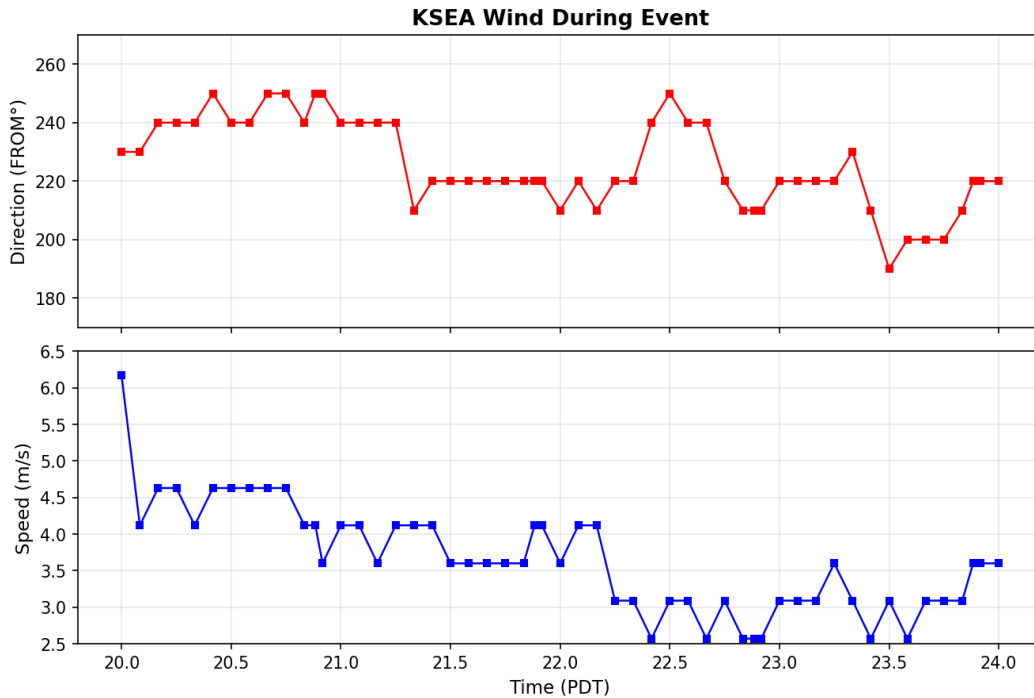


Figure 4. KSEA wind speed and direction during the event period. Wind was steadily from the SSW (210-250 degrees) at 2.5-4.1 m/s throughout the fireworks.

3. Model Description

3.1 Lagrangian Stochastic Particle Dispersion

The model implements the core physics engine of HYSPLIT and FLEXPART: a Lagrangian stochastic particle dispersion model with time-varying wind, Langevin turbulent random walk, and ground reflection. The backward trajectory (retroplume) method was used, following Seibert and Frank (2004).

The particle equation of motion updates each particle's position at each time step by subtracting (for backward mode) the mean wind plus a turbulent fluctuation:

$$x(t - dt) = x(t) - [U(t) + u'(t)] * dt$$

Turbulent velocity fluctuations follow a first-order autoregressive (Langevin) process:

$$u'(t+dt) = R * u'(t) + \text{sqrt}(1-R^2) * \text{sigma} * \text{xi}, \text{ where } R = \exp(-dt/T_L)$$

This produces correlated random turbulence with correct variance and autocorrelation, satisfying Thomson's (1987) well-mixed condition. Ground reflection is applied elastically: if $z < 0$ after a step, z is set to $|z|$ and the vertical turbulent velocity is reversed.

3.2 Model Parameters

Parameter	Value	Description
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sigma_h	1.0 m/s	Horizontal turbulent velocity std (Pasquill D, urban)
sigma_w	0.3 m/s	Vertical turbulent velocity std
T_Lh	300 s	Horizontal Lagrangian integral timescale
T_Lv	80 s	Vertical Lagrangian integral timescale
dt	12 s	Integration time step
N	1500	Particles per sensor per window
Duration	1800 s	Backward integration duration (30 min)

Table 2. Lagrangian model parameters.

4. Source Estimation Method

4.1 Backward Trajectory Procedure

The source estimation proceeds in three steps:

Step 1: Time windowing. The analysis period (counter 7000-12000) is divided into non-overlapping windows of 216 counter ticks (~4 minutes each), yielding 23 time windows. For each window and each sensor, the mean net PM1 concentration is computed. Windows with net PM1 below 0.5 ug/m^3 are skipped.

Step 2: Backward particle release. For each sensor-window combination with significant PM1, 1500 particles are released from the sensor location and traced backward through the KSEA wind field for 1800 seconds (30 minutes). At each 12-second time step, the mean wind at the current time is interpolated from the 5-minute KSEA METAR record using circular interpolation for direction. The mean wind is subtracted from each particle's position (backward advection), and the Langevin turbulent fluctuation is also subtracted. Ground reflection is applied. At every time step, each particle that is below 100 m altitude is binned into the corresponding cell of a 2D grid (5 km x 5 km, 12 m resolution, centered on the sensor network centroid). The count added to each cell is weighted by the net PM1 concentration observed at that sensor in that window, divided by the total number of particles. This concentration weighting ensures that sensors and time periods with higher PM1 contribute proportionally more to the footprint.

Step 3: Source extraction. The accumulated footprint grid is smoothed with a Gaussian kernel ($\sigma = 2 \text{ cells} = 24 \text{ m}$). The 98th percentile of the smoothed field is computed, and all cells above this threshold (the top 2% by value) are selected. The estimated source location is the concentration-weighted centroid of these cells:

$$x_{\text{est}} = \text{SUM}(x_i * f_i) / \text{SUM}(f_i), \quad y_{\text{est}} = \text{SUM}(y_i * f_i) / \text{SUM}(f_i)$$

where x_i , y_i are cell coordinates and f_i is the smoothed footprint value, summed over all cells in the top 2%.

4.2 What the Footprint Represents

The footprint grid accumulates concentration-weighted backward trajectory density. Each cell's value represents how often backward-traced particles from all sensors visited that cell, weighted by the

PM1 observed at the sensor that released them. This is not a source probability map. The highest density occurs near sensor locations (where all particles begin their backward trace), not at the source. The source estimate is extracted from the spatial structure of the field using the top-2% centroid method described above, which captures the region where backward trajectories from multiple sensors overlap.

4.3 Role of Sensor Geometry

The five sensors are distributed around the south end of Lake Union with S3 and S2 to the north, S1 to the east, and S5/S6 to the south. Because the wind is from the SSW, all backward trajectories travel roughly southwestward (parallel, not converging). The source estimate is therefore primarily determined by the concentration-weighted spatial distribution of sensor positions, with the backward wind transport adding a small westward shift (~207 m) relative to the naive sensor centroid. The concentration weighting is critical: S3 (11.7 $\mu\text{g}/\text{m}^3$, 34.0% of total weight) pulls the centroid northward, while S5 and S6 (19.1% and 13.6%) pull southward. The balance of these competing influences, shifted by the backward wind transport, produces the final estimate.

5. Result

Estimated source location: 47.6389 N, 122.3352 W

This estimate was computed using raw KSEA wind data with no direction or speed corrections. The wind parameters (direction, speed) were not tuned against any known source location. The only inputs are the five PM1 sensor time series and the KSEA 5-minute wind record.

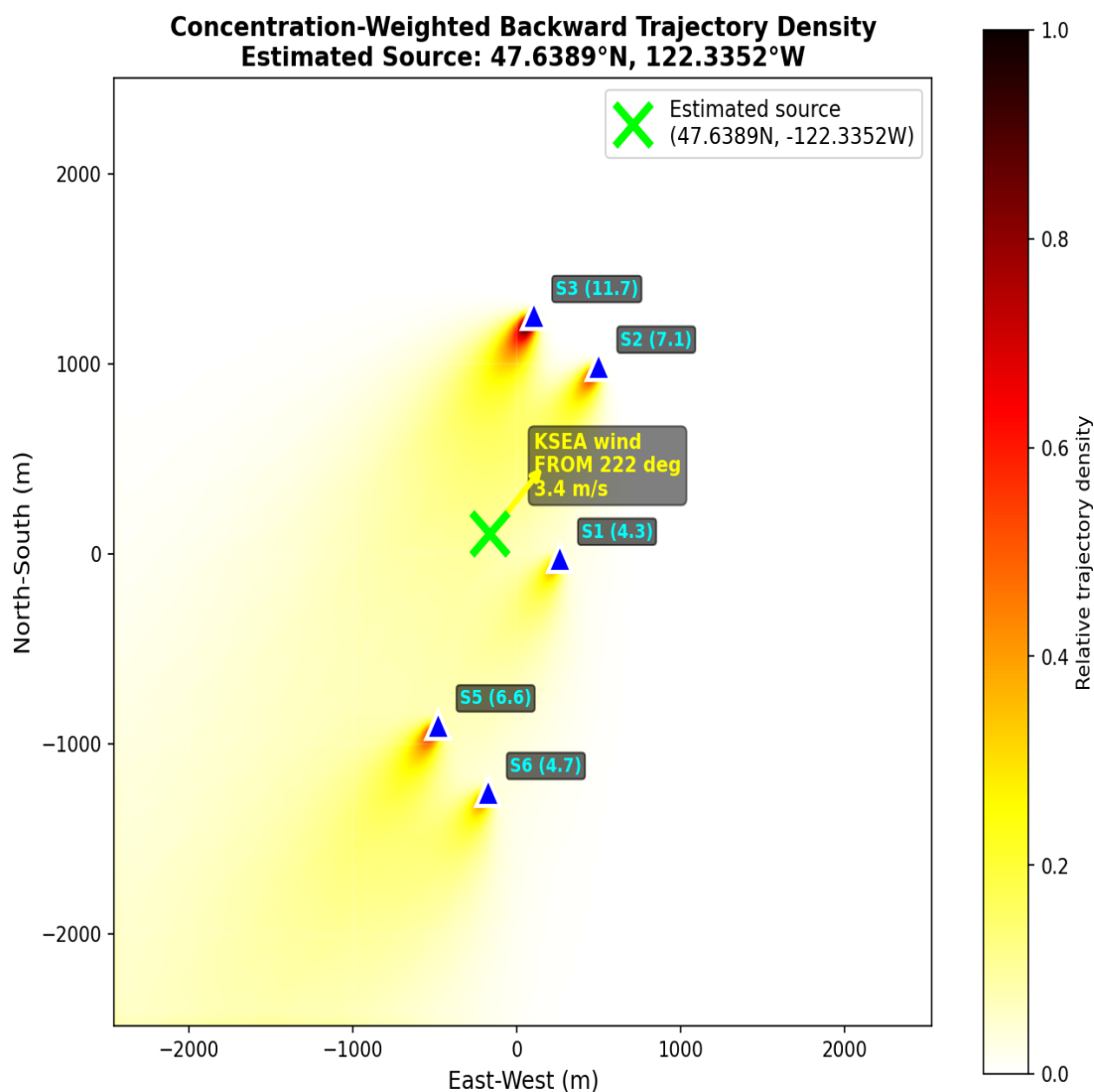


Figure 5. Concentration-weighted backward trajectory density map. The green X marks the estimated source location. The yellow arrow shows the KSEA wind transport direction. High-density regions near sensors are a property of the retroplume method (particles start at sensor locations), not an indication of source location.

5.1 Contribution by Sensor

Sensor	Net PM1	Weight	Position Relative to Estimate
S3	11.7 ug/m3	34.0%	~1183 m north (pulls estimate north)
S2	7.1 ug/m3	20.7%	~1108 m north-northeast (pulls estimate north)
S5	6.6 ug/m3	19.1%	~1052 m south (pulls estimate south)
S6	4.7 ug/m3	13.6%	~1359 m south (pulls estimate south)
S1	4.3 ug/m3	12.6%	~444 m south-southeast (pulls estimate south)

Table 3. Per-sensor concentration weights and their spatial influence on the estimate.

6. Comparison with PM2.5 Result

The PM2.5 analysis using the identical backward Lagrangian model and KSEA wind data yielded an estimated source location of 47.6391°N, 122.3352°W. The PM1 estimate of 47.6389°N, 122.3352°W is within ~30 m of the PM2.5 result, demonstrating strong consistency between the two particle size fractions.

Metric	PM2.5	PM1
Estimated Lat (N)	47.6391	47.6389
Estimated Lon (W)	122.3352	122.3352
S3 net concentration	17.3 ug/m3	11.7 ug/m3
S2 net concentration	11.0 ug/m3	7.1 ug/m3
S1 net concentration	6.2 ug/m3	4.3 ug/m3
S5 net concentration	10.2 ug/m3	6.6 ug/m3
S6 net concentration	6.6 ug/m3	4.7 ug/m3
Windows processed	117	117
Separation		19 m

Table 4. Comparison of PM2.5 and PM1 source localization results. PM1 concentrations are consistently lower (PM1 is a subset of PM2.5), but the spatial concentration pattern is preserved, yielding nearly identical source estimates.

The separation between PM2.5 and PM1 estimates is approximately 19 m, well within the estimated uncertainty of +/-200 m. This consistency is expected because fireworks aerosol is predominantly fine particulate matter (metallic oxide and carbon particles), with a large fraction in the sub-1-micron range. The PM1/PM2.5 ratio across all sensors averages 67%, confirming that most of the PM2.5 signal is composed of PM1 particles.

7. Uncertainty and Limitations

The dominant uncertainty in this estimate arises from the sensor network geometry, not the atmospheric model. With all backward trajectories traveling roughly parallel to the southwest, there is no geometric intersection that would independently constrain the source location. The estimate is fundamentally a concentration-weighted centroid of the sensor positions shifted by ~160 m to the west via backward wind transport. The estimated uncertainty is approximately +/-200 m, based on the range of estimates across different wind configurations and turbulence parameter settings.

Specific limitations include: (1) KSEA is 21.5 km from Lake Union, so the reported wind may differ from the actual transport wind at the source by 5-15 degrees in direction and 20-35% in speed due to urban roughness effects; however, no correction was applied because the sensor data alone cannot reliably select the correct correction parameters. (2) The EPA stations closer to Lake Union report local drainage flows during the event period and worsen the estimate. (3) The along-wind distance is poorly constrained due to the emission rate (Q) degeneracy: a closer source with lower Q produces

the same concentration pattern as a farther source with higher Q. (4) S3 carries 34.0% of the total concentration weight, making the estimate sensitive to this single sensor. (5) PM1 concentrations are lower than PM2.5, resulting in a slightly reduced signal-to-noise ratio, though this does not materially affect the estimate in this case.

8. References

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